

# Tanvir Ahmed

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## RESEARCH INTERESTS

Cyber Security • Large Language and Vision–Language Models • Quantum Machine Learning • Computer Vision

## RESEARCH SUMMARY

Computer science graduate and independent researcher in machine learning, working mainly on security for networked systems. Four of my papers were accepted in 2026. The first-author one, at ICTAI, shows that feature selection can leave overall accuracy almost unchanged while making a rare attack class far harder to detect, and gives a method that finds the harmed classes and repairs the subset. A related line treats intrusion-detection feature selection as a combinatorial optimization problem, solved with QAOA on IBM Quantum hardware and with simulated annealing, then benchmarked against classical selectors on industrial IoT traffic (X-IIoTID, NF-ToN-IoT, NF-UNSW-NB15). I also co-authored an accepted agent that pairs a vision–language model with an LLM and runs both locally, so web pages are screened on consumer hardware without anything leaving the device, and two accepted papers on trustworthy tuberculosis microscopy. Most of my effort goes into evaluation that holds up: patient-disjoint splits, leakage audits and per-class results rather than a single headline score. Code is public for every project.

## EDUCATION

Independent University, Bangladesh

January 2015 — December 2019

Department of Computer Science and Engineering

Bachelor of Science in Computer Science, Minor in Environmental Management

CGPA: 2.92 out of 4.00 (Final-year GPA: 3.07)

## PUBLICATIONS

### Conference papers

Accepted

- **Tanvir Ahmed**, Jannatul Ferdous Shorna, Md Mahbubur Akash, Md Khorshed Alam, and Md Tarek Hasan. “DARE-FS: A Selector-Agnostic Operator for Detecting and Repairing Class-Specific Feature-Selection Harm.” *2026 IEEE 38th International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026. Code: [GitHub].
- Md Faisal Sheikh, **Tanvir Ahmed**, Enam Ahmed Taufik, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Beyond Confidence: A Leakage-Audited Selective-Prediction Study of Acid-Fast Bacilli Detection in Ziehl–Neelsen Microscopy.” *2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2026. Code: [GitHub].
- Swagotam Malakar, **Tanvir Ahmed**, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Towards Trustworthy Tuberculosis Detection: Subgroup Calibration of Object Detectors in Sputum-Smear Microscopy.” *2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2026. Code: [GitHub].
- Dhiman Kumar Sarker, Fyaz Nafin Rahman, Md Rezaul Karim Khan, Sushmoy Paul Rohit, **Tanvir Ahmed**, and Md Mizanur Rahman. “Real-Time Browser Threat Detection Security Agent via Local VLM–LLM.” *2026 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON)*, 2026. Code: [GitHub].

### Journal paper

Under Review

- Abhijit Kumar Ghosh, Ar Rafiul Islam Rafim, Fahmid Al Farid, **Tanvir Ahmed**, Enam Ahmed Taufik, Jia Uddin, and Hezerul Abdul Karim. “Gated Dual-Scale Feature Fusion for Colorectal Cancer Grading in Centralized and Federated Pathology.” *Informatics in Medicine Unlocked* (Elsevier), 2026. Code: [GitHub].

## PROJECTS

### QCIDS: Quantum–Classical Feature Selection for Network Intrusion Detection [GitHub]

- Cast intrusion-detection feature selection as a QUBO solved by QAOA ( $p = 1, 2, 3$ ) on simulators and IBM 156-qubit hardware, by simulated annealing, and by classical baselines under matched leakage-free cross-validation; at  $k = 15$  on NF-ToN-IoT, QAOA  $p = 2$  held macro-F1 at 0.763 where greedy mutual information fell from 0.760 to 0.718.

### CovNCG: Training-Free Dense Quantum Feature Maps [GitHub]

- Designed a training-free quantum kernel from non-commuting generator blocks (three features per two qubits) with a closed-form kernel-variance result scaling as  $c^4$ , which turns avoidance of exponential concentration into a bandwidth tuning rule; evaluated with a projected kernel and SVM on five medical datasets, with IBM QPU validation.

***Q-PlasmaNet: Hybrid Classical–Quantum Classifier for Plasma Cell Recognition*** [GitHub]

- Paired a frozen CNN backbone with a 4–8 qubit data re-uploading variational quantum head (PennyLane) at matched parameter budget, reaching F1 0.927 with 178 trainable parameters — statistically tied with a parameter-matched MLP (0.932) and below a 2.2M-parameter MobileNetV2 (0.949), establishing parity rather than quantum advantage.

***PCMMD: Federated Learning and Knowledge Distillation for Plasma Cell Detection*** [GitHub]

- Benchmarked centralized, few-shot, knowledge-distillation (EfficientNet-B0 to MobileNetV3) and federated (FedAvg/FedProx, one client per patient) training for plasma-cell classification of bone-marrow crops, under patient-disjoint folds and image-hash de-duplication.

***Direct-WAM: Self-Supervised Denoising for Two-Photon Calcium Imaging*** [GitHub]

- Proved a bias-floor theory of self-supervised objectives (per-objective error floor, ensemble bias-crossing, distillation transfer bound) and produced a corrected AI4Life CIDC25 benchmark in which a faithfully retrained DeepCAD-RT reaches +23.65 dB stSNR, while the  $F_0$ -free Direct-WAM 3D U-Net distilled from a 13-way ensemble reaches +18.15 dB.

**RESEARCH EXPERIENCE****Independent Researcher**

Boca Raton, Florida, USA

*Independent and collaborative research*

January 2025 — Present

**Research collaborators:** Prof. Jia Uddin (Woosong University, South Korea); Dr. Mohammad Zavid Parvez (RMIT University, Australia); Assoc. Prof. Subrata Chakraborty (University of New England, Australia); Prof. Prabal Datta Barua (University of Southern Queensland, Australia)

- Author and co-author five papers with international collaborators: four accepted at IEEE and DICTA venues in 2026, one first-author, and one journal manuscript under review.
- Run the full experimental pipeline on each project: auditing the data for leakage, building and training the models, benchmarking against classical and quantum baselines, and reporting results over multiple seeds with confidence intervals.

**PROFESSIONAL EXPERIENCE****Career Break — Relocation to the United States**

Boca Raton, Florida, USA

*F-2 dependent status*

January 2023 — December 2024

- Relocated from Bangladesh to the United States as an F-2 dependent, a status that does not authorize employment; used the period for self-directed study in machine learning, which led into the independent research work above.

**Upwork & Fiverr**

Remote — Dhaka, Bangladesh

*Freelance Web Developer and E-commerce Specialist*

December 2014 — November 2022

- Delivered e-commerce storefronts for international clients, both as customized Shopify builds (themes, product catalogues, checkout flows) and as custom-developed websites written from scratch to client specification.
- Ran day-to-day store operations across order and inventory management, customer inquiries, and issue resolution, sustaining repeat client relationships over an eight-year freelance practice.

**IDOL Focus — A Professional Training Institute**

Dhaka, Bangladesh

*Web Developer Intern*

September 2019 — December 2019

**Supervisors:** Md. Asif Bin Khaled, Lecturer, Department of Computer Science and Engineering, IUB (academic); Md. Al Mamun, IDOL Focus (industry)

- Built the Admin and Agent modules of an online job portal (*idolfocusjobs.com*) serving four user roles: role-based authentication, jobseeker and employer account management, CV database administration, job-posting approval, and payment handling.
- Implemented in PHP, JavaScript, Ajax, jQuery, and Bootstrap over MySQL; documented in an undergraduate internship report submitted toward the BSc degree.

**TECHNICAL SKILLS**

|                        |                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Security:</b>       | Network intrusion detection, IIoT traffic analysis, threat detection agents, data-leakage auditing     |
| <b>Trustworthy ML:</b> | Calibration (ECE, Brier), selective prediction, subgroup and worst-group evaluation, object detection  |
| <b>LLM &amp; VLM:</b>  | Local LLM and vision–language model inference, 4-bit quantized deployment, multimodal threat detection |
| <b>Programming:</b>    | Python, SQL, PHP, JavaScript                                                                           |
| <b>ML Frameworks:</b>  | PyTorch, Scikit-learn, LightGBM, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib                         |
| <b>Quantum ML:</b>     | PennyLane, Qiskit, QAOA/QUBO formulation, variational circuits, quantum kernels, IBM Quantum hardware  |
| <b>Core Areas:</b>     | Machine Learning, Computer Vision, Deep Learning, Federated Learning, Knowledge Distillation           |
| <b>Databases:</b>      | MySQL, phpMyAdmin                                                                                      |
| <b>Web / Software:</b> | HTML, CSS, JavaScript, jQuery, Ajax, Bootstrap, Shopify, Apache Web Server                             |
| <b>Tools:</b>          | L <sup>A</sup> T <sub>E</sub> X, Git, Jupyter Notebook, Google Colab                                   |
| <b>Languages:</b>      | Bangla (native), English (fluent)                                                                      |